

Introduction:

Low-carbon commuting modes are recognized as important strategies for reducing greenhouse gas emissions in urban areas. To such models belong, for example, cycling, walking, and public transportation. The adoption of such behaviors is important and depends not only on infrastructure and policy incentives but also on social dynamics. Previous studies suggest that awareness campaigns, social norms, and peer interactions can influence people's decisions to adopt more sustainable commuting practices. Social networks could shape how these behaviors spread, because individuals are often influenced by the actions and attitudes of people around them.

There is some research on this topic. This existing research shows us the importance of social influence in promoting sustainable behavior, emphasizing that network structure and interpersonal influence can affect both the speed and scale of behavioral diffusion. There is a suggestion that interventions targeting strategically positioned individuals, especially ones with high network centrality, could encourage broader behavioral adoption more effectively than untargeted approaches.

It is important to mention that these factors are often examined separately, which leaves limited understanding of how social influence, network structure, and intervention strategies interact within a single modeling framework. Specifically, not enough attention has been given to how different approaches to selecting initial adopters shape the diffusion of low-carbon commuting behavior under varying adoption conditions. There are different approaches, such as random selection or targeting highly connected individuals.

To address this issue, this study will apply an agent-based simulation to model the diffusion of low-carbon commuting behavior within a synthetic social network. Two intervention strategies are compared: random selection of initial adopters and centrality-based targeting of highly connected individuals. By evaluating these strategies across different adoption thresholds, the study investigates whether targeting central actors accelerates diffusion and increases overall adoption, offering insights for the design of more effective urban sustainability interventions.

Methodology:

For this project, we used an agent-based simulation model to study how low-carbon commuting can spread through a social network. In the model, each person is represented as an agent, and the connections between people represent social relationships. The main aim of the model is to compare two different ways of starting the adoption process: choosing initial adopters randomly and choosing the most central or well-connected individuals.

The selected model is a threshold-based diffusion model. This means that a person does not adopt low-carbon commuting automatically. Instead, they adopt only when enough of their connected neighbors have already adopted. This is useful for our project because commuting behavior can be influenced by social pressure, peer behavior, and what people see others around them doing. Low-carbon commuting in this model can represent behaviors such as walking, cycling, using public transport, car sharing, or carpooling instead of using a private car.

The social network was created using Python and the NetworkX library. We used a Barabasi–Albert scale-free network. This type of network was chosen because it contains some highly

connected individuals, while most people have fewer connections. This fits our research question because we are testing whether targeting these highly connected individuals leads to more adoption than choosing people randomly. In the model, the network contains 1,000 agents. The parameter $m = 3$ was used, meaning that each new node connects to three existing nodes when the network is generated.

Each agent in the model has one of two states. A value of 0 means that the agent has not adopted low-carbon commuting, while a value of 1 means that the agent has adopted it. At the beginning of each simulation, 5% of the agents are selected as initial adopters. Since the model has 1,000 agents, this means that 50 agents start as adopters in each run.

Two targeting strategies were tested. In the random targeting strategy, the 50 initial adopters are selected randomly from the whole network. In the central targeting strategy, the 50 agents with the highest number of connections are selected as the initial adopters. These agents are considered central because they are connected to more people and therefore may have more influence in the network.

The adoption rule is based on a threshold. At every time step, each non-adopting agent checks how many of its neighbors have already adopted low-carbon commuting. If the proportion of adopted neighbors is equal to or greater than the threshold, that agent also adopts. For example, if the threshold is 0.3, an agent adopts when at least 30% of its neighbors have already adopted. Once an agent adopts, it stays as an adopter for the rest of the simulation.

We tested five different threshold values: 0.1, 0.2, 0.3, 0.4, and 0.5. These values represent different levels of social influence needed for adoption. A low threshold, such as 0.1, means people are easier to influence because only 10% of their neighbors need to adopt first. A higher threshold, such as 0.5, means adoption is harder because at least half of the person's neighbors must already be adopters.

Each simulation was run for a maximum of 50 time steps. However, the simulation stops earlier if no new agents adopt during a time step. This stopping condition shows that the spread of adoption has ended. To make the results more reliable, each condition was repeated 100 times. This was important because random targeting and network generation can produce different outcomes in different runs.

No real commuting data was collected for this project. Instead, the model uses synthetic data created through the simulation. This means that the agents and their connections are not real people, but simulated individuals in an artificial network. This approach was suitable because the purpose of the project was not to predict exact real-world commuting behavior, but to test how different targeting strategies can affect adoption in a controlled model. It also avoids privacy issues because no personal travel or social network data is used.

The main outcome measured was the final adoption rate, which shows the proportion of agents who adopted low-carbon commuting by the end of the simulation. We also recorded adoption over time, so we could see how quickly adoption spread in each strategy. In addition, we compared the average final adoption rate for random and central targeting at each threshold level. The standard deviation was also calculated to show how much the results varied across the 100 repeated simulations.

We also measured cascade probability. In this project, a large cascade was defined as a simulation where more than 50% of the agents adopted low-carbon commuting. This helped us understand not only which strategy had a higher average adoption rate, but also which strategy was more reliable in creating widespread adoption.

Overall, the methodology allowed us to compare random targeting and central targeting under the same network conditions. By changing the threshold values and repeating the simulations many times, we could observe how the effectiveness of each strategy changed when adoption became easier or harder.

Results:

After running the model that was created for this task, we got results, which we would share in this section. The simulation results showed clear differences between random targeting and central targeting strategies, particularly as adoption thresholds increased. At lower threshold values (0.1 and 0.2), both strategies resulted in full adoption of low-carbon commuting behavior. In these scenarios, the average final adoption rate reached 100% for both approaches, indicating that when individuals require only limited social influence before adopting, the choice of targeting strategy has little impact on the outcome.

To see the differences between the two strategies, we need higher threshold levels. At a threshold of 0.3, central targeting continued to produce complete adoption. The average final adoption rate is 100%. Random targeting achieved only approximately 33.6% adoption. This suggests that as adoption becomes more difficult and individuals require stronger social reinforcement, the position of initial adopters within the network becomes increasingly important.

The contrast became even stronger at thresholds of 0.4 and 0.5. Under random targeting, adoption remained close to the initial 5% seed group, reaching average final adoption rates of approximately 5.8% and 5.6%, respectively. In comparison, central targeting maintained very high levels of adoption, reaching 100% at a threshold of 0.4 and approximately 98.7% at a threshold of 0.5. These findings indicate that targeting highly connected individuals remained effective even under more demanding adoption conditions.

Table 1. Average final adoption rate by threshold and targeting strategy

Threshold	Random targeting	Central targeting	Central - random
0.1	1.000	1.000	0.000
0.2	1.000	1.000	0.000
0.3	0.336	1.000	0.664
0.4	0.059	1.000	0.941
0.5	0.056	0.982	0.927

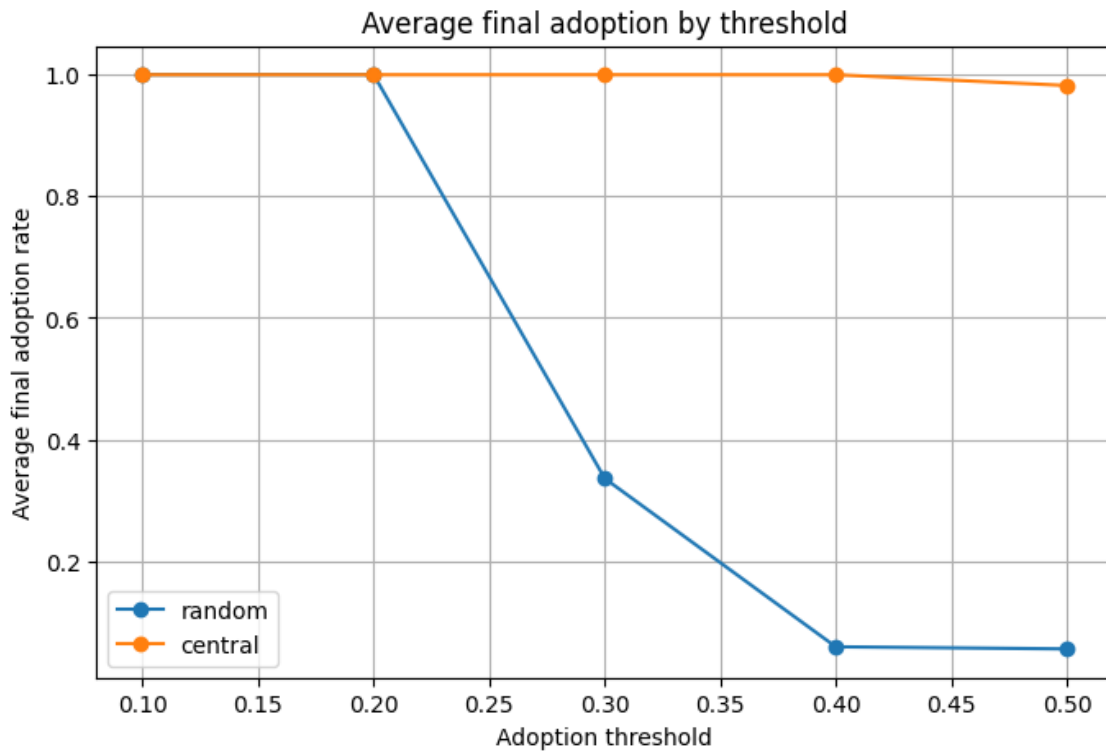


Figure 1. Average final adoption rate by threshold and targeting strategy

The speed of adoption also differed substantially between strategies (Figure 2). At lower thresholds, adoption spread rapidly regardless of the targeting approach. However, as threshold values increased, central targeting consistently produced faster diffusion than random targeting. At a threshold of 0.3, adoption spread quickly and reached near-complete diffusion within a small number of time steps when central individuals were targeted, while random targeting resulted in slower and more limited spread. At thresholds of 0.4 and 0.5, diffusion under random targeting largely stalled, whereas central targeting continued to generate widespread adoption.

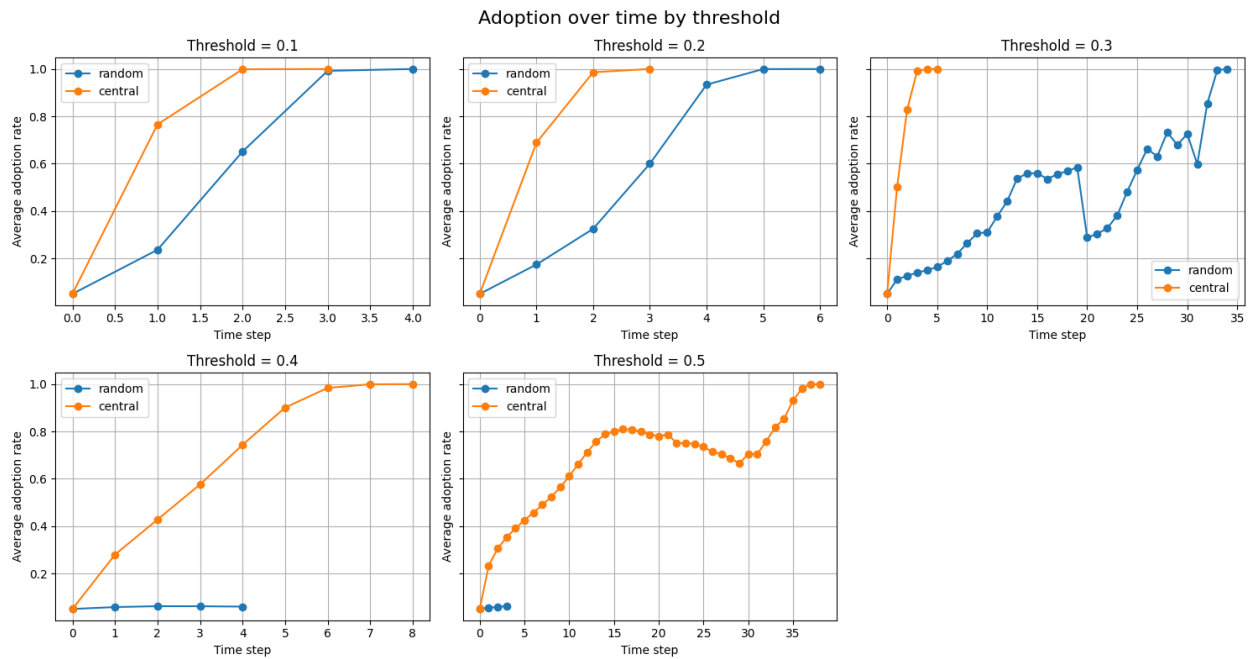


Figure 2. Adoption over time by threshold and targeting strategy

In Figure 2, some random-targeting lines stop before the maximum of 50 time steps. This happens because the simulation stops when no new agents adopt, meaning the diffusion process has reached a stable point rather than that data are missing.

The repeated simulations further revealed differences in reliability between the two strategies. Results for random targeting showed greater variation between simulation runs, particularly at the threshold of 0.3, where some runs produced relatively large cascades while others resulted in limited adoption. In contrast, central targeting produced highly consistent outcomes with very little variation across simulations. This suggests that central targeting was not only more effective on average but also more stable in generating widespread adoption.

To further assess diffusion success, cascade probability was measured. A large cascade was defined as a simulation in which more than 50% of agents adopted low-carbon commuting behavior. At thresholds of 0.1 and 0.2, both strategies achieved a cascade probability of 100%. At a threshold of 0.3, central targeting continued to achieve large cascades in all simulations, while random targeting reached this outcome in only 23% of cases. At thresholds of 0.4 and 0.5, random targeting no longer produced large cascades, whereas central targeting maintained cascade probabilities close to 100%.

Table 2. Cascade probability by threshold and targeting strategy

Threshold	Random targeting	Central targeting
0.1	1.00	1.00
0.2	1.00	1.00
0.3	0.23	1.00
0.4	0.00	1.00
0.5	0.00	0.97

Table 2 shows that central targeting was more reliable in producing large cascades. At low thresholds, both strategies produced large cascades in all simulations. When the threshold increased, random targeting became much less reliable, while central targeting still produced large cascades in almost all runs.

Overall, the findings support the hypothesis that targeting highly connected individuals increases both the speed and extent of low-carbon commuting adoption compared to random targeting. The advantage of central targeting became especially pronounced when behavioral adoption required stronger social influence.

Discussion

Our results mostly matched what we expected based on the papers we read before starting the project. The clearest connection is with the Granovetter paper from 1978 about threshold models. In that paper he explained that when people need a certain number of others to act before they act themselves, small differences in who starts the process can lead to very different outcomes for the whole group. We saw exactly this in our simulation. When the threshold was low, like 0.1 or 0.2, it did not really matter who the initial adopters were because the behavior spread easily no matter what. But once we raised the threshold to 0.3 and higher, the choice of starting people made a huge difference. Central targeting kept giving full adoption while random targeting basically failed and stayed close to the starting 5 percent. This makes sense if you think about it because when someone needs 30 or 40 percent of their neighbors to adopt first, it really helps if those neighbors are already connected to many other people. There was also a Dutch study we read that used a threshold based agent model with 1000 agents and 100 simulation runs, which is very similar to what we did. That study found that behaviors spread better when they are visible to others around you, and we think this also connects to our results because central people are seen by more people in the network so when they adopt it triggers more others to adopt as well.

One thing that was really interesting when we compared our results to the literature is that our simulation actually gives the opposite result from a real world study we reviewed. The Lancet study carried out a real experiment in villages in Honduras where they tested central targeting against random targeting to spread health behaviors. In that study central targeting did the worst out of all three strategies they tested and random targeting performed better. The researchers explained this probably happened because in real social networks the most connected people tend to already know each other, so the behavior spreads quickly between the hubs but then gets stuck and never really reaches the rest of the population. In our simulation this did not happen at all and central targeting won very clearly at every high threshold. We think the reason for this difference is that our network was artificially generated and probably do not have the same kind of clustering that real social networks have in actual communities. This is an important point because it shows that what happens in a simulation and what happens in real life can be quite different sometimes. Our model is good for showing the theory but we cannot say from our results alone that central targeting will always work better in practice.

Our findings also connect to the papers we read about how peer influence affects commuting choices in general. One study looked at employees at a university in Grenoble in France and found that when more of your coworkers use active transport like cycling or walking, the chance that you also start doing the same goes up by a lot, in some cases by up to 70 percent points. This is a strong evidence that the people around you really do affect how you decide to commute, and this is exactly the core idea that our whole model is built on. Another paper we reviewed tested whether sending social messages to people could push them toward choosing low-carbon travel. They found the messages helped a little but the effect was quite weak on its own and the researchers said other things like better infrastructure or financial support are also needed. We think this is relevant to our results because even though our simulation shows that central targeting can create a big wave of adoption through the network, in real life people also need practical things to actually change how they travel every day. So the social influence part that we simulated is real and does matter, but based on what the literature says it is probably only one part of what is needed to actually get people to switch to low-carbon commuting in a real city.

Limitations:

One of the biggest limitations in our project is that we did not use any real data at all. The whole network, all the people and all the connections between them, was generated by the computer using the Barabasi–Albert algorithm. A scale-free network does give you some of the properties that real networks have, like a few very connected people and many less connected ones, but it is still not the same as how people are actually connected in real life. Real social connections between commuters depend on things like where someone lives, what kind of job they have, how long they have been in a city, and who they happen to know from different parts of their life. None of this could be included in our model. When we were doing our literature review we read a study about employees at a university in Grenoble where researchers actually surveyed real people and collected real data about their commuting habits and their workplace connections. That kind of study can capture things that a simulation simply cannot. So while our results are useful for understanding the theory behind targeting strategies, they do not tell us what would actually happen if someone tried to use these strategies in a real city or organization.

Another limitation is the way we modeled the individual people in our simulation. Every single agent in the model uses the exact same threshold, which means everyone needs the same share of their neighbors to have adopted before they adopt themselves. But in real life people are very different from each other. One person might start cycling to work after just two colleagues do it, while another person might never switch even if almost everyone around them already did, maybe because of where they live or because there is no good bus connection near their house. Granovetter talked about this in his 1978 paper and said that the differences in thresholds across a group is what actually shapes the final outcome for the whole group. So using one fixed threshold for everyone in the network is a big simplification and it probably has some effect on our results. On top of that, in our model once someone adopts low-carbon commuting

they stay that way until the end of the simulation and never go back to their old behavior. This is obviously not how it works in real life because people change their habits depending on weather, personal situation, or if their circumstances change. We also did not include any real world factors like how far someone lives from their workplace, how much different transport options cost, or whether there are safe cycling paths available. These kind of factors clearly have a big effect on commuting in real life but we had no way to include them in our model.

The last limitation we want to mention is that the network in our simulation was fully static the whole time. The connections between agents was always the same from the very beginning to the very end, they never changed. But real social networks change all the time because people change jobs, move to different neighborhoods, meet new people, or lose contact with old ones. A better version of this model would allow the network to slowly change as the simulation is running. We also only compared two targeting strategies in this project, random and central, which is another limitation. The Lancet study from Honduras that we reviewed for our literature actually compared three strategies and the third one they called friendship nomination, where you randomly pick some people and ask each of them to name one friend, and then you target those friends as the initial adopters. They found that this third strategy actually worked better than both central and random targeting for one of the behaviors they tested. Not including this strategy in our simulation made our comparison less complete than it could have been. For future work it would be good to add friendship nomination as a third strategy so the comparison is richer, to give different agents different thresholds so the model is more realistic, to let people stop using low-carbon commuting and go back to old habits, and if possible to test the model using a real social network from an actual city instead of a generated one. These additions would bring the model much closer to real life and would make the results more useful for anyone who wants to design a real intervention to promote low-carbon commuting.

Conclusion:

To conclude, we used an agent-based social network simulation to study how low-carbon commuting behaviour can spread through peer influence. The main research question was whether targeting central individuals leads to higher adoption of low-carbon commuting compared to targeting random initial adopters. Based on the simulation results, central targeting was generally more effective than random targeting, especially when the adoption threshold was higher.

When the threshold was low, such as 0.1 and 0.2, both random targeting and central targeting resulted in full adoption. This means that when people are easy to influence, the choice of targeting strategy does not make a big difference because adoption spreads easily through the network anyway. However, the results changed when the threshold became higher.

At a threshold of 0.3, central targeting produced an average final adoption rate of 1.0, while random targeting only produced an average final adoption rate of about 0.336. This shows a clear difference between the two strategies. The same pattern became even stronger at thresholds of 0.4 and 0.5. At threshold 0.4, central targeting still reached full adoption, while

random targeting only reached about 0.059. At threshold 0.5, central targeting reached about 0.982, while random targeting stayed very low at about 0.056.

The cascade results also support this conclusion. A large cascade was defined as more than 50% of agents adopting low-carbon commuting. For central targeting, the probability of a large cascade was 1.0 for thresholds from 0.1 to 0.4, and 0.97 at threshold 0.5. For random targeting, large cascades happened at low thresholds, but became much less likely when the threshold increased. At threshold 0.3, random targeting only produced a large cascade in 23% of runs, and at thresholds 0.4 and 0.5 it produced no large cascades.

Overall, the results answer the research question clearly. Targeting central individuals does lead to higher adoption of low-carbon commuting than targeting random individuals in this simulation, mainly when adoption requires stronger social influence. Central individuals are more connected in the network, so when they adopt early, they expose more people to the behavior and help the adoption spread further. However, the results also show that targeting strategy matters less when the adoption threshold is very low, because adoption spreads easily in both cases.

Therefore, the main conclusion of this project is that central targeting is a better strategy for spreading low-carbon commuting in a social network model, especially when people need a stronger level of peer influence before changing their behavior.

References

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